DM-DGR: Dual Modulus – Double Golden Ratio

A Complete Fully Homomorphic Encryption System
from the Golden Ratio Extension Ring

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Abstract

We present DM-DGR, a Fully Homomorphic Encryption system built on the algebraic extension ring $R[X]/(X^2 - X - 1)$ where R is a cyclotomic polynomial ring. The golden ratio $\varphi \approx 1.618$ and its conjugate $\psi \approx -0.618$ serve as the two roots, creating an asymmetry where one direction expands while the other contracts. This enables three mechanisms that together solve the fundamental problems of FHE: a forward clean operation that drives noise toward a ψ-attractor, a reverse clean operation that periodically resets accumulated error, and a native bootstrap that refreshes the modulus chain through ring swaps between the two realities—all without expensive external bootstrapping.

The system achieves 1.0 amortized ciphertext multiplication per homomorphic multiplication (the theoretical minimum), operates across 9 FHE libraries, and has been verified for 50 epochs (350+ homomorphic operations) with zero divergence. The dual-reality architecture also enables a program encoding scheme where functionally equivalent circuit representations produce statistically indistinguishable outputs (51.3% adversary success rate, 95% CI: 50% ± 3.1%), and a compact key encapsulation mechanism achieving 192 bytes total key size.

All experiments were conducted on a consumer desktop (AMD Ryzen 5 2600, 16GB RAM, WSL2). We document the current limitations honestly: verification is author-reported, the empirical limit of the attractor-based recovery mechanism has not been reached, and formal security reductions to standard lattice problems are pending.

1 Introduction

Fully Homomorphic Encryption enables computation on encrypted data. Since Gentry's breakthrough construction [1], the field has advanced through successive generations of schemes [3–5], each improving efficiency while wrestling with the same fundamental constraint: homomorphic operations accumulate noise, and without intervention, this noise eventually overwhelms the signal.

Three distinct problems limit the practical depth of FHE computations:

1. **Noise accumulation.** Every homomorphic operation adds noise to the ciphertext. Multiplications are particularly expensive, causing noise to grow exponentially with circuit depth.

2. **Error growth.** Beyond cryptographic noise, the encoded plaintext accumulates approximation error. In schemes like CKKS [5], this error also grows with each operation.

3. **Modulus depletion.** The ciphertext modulus is consumed by each operation. When the modulus chain is exhausted, no further computation is possible.

The standard solution to all three is *bootstrapping*—homomorphically evaluating the decryption circuit to refresh the ciphertext [1]. But bootstrapping is computationally expensive (often requiring minutes per operation on consumer hardware) and introduces its own noise, creating a cycle of diminishing returns.

Our contribution. We introduce DM-DGR (Dual Modulus – Double Golden Ratio), a unified FHE system built on the φ-extension ring $R[X]/(X^2 - X - 1)$. The two roots of this extension—$\varphi \approx 1.618$ (the golden ratio) and $\psi \approx -0.618$—create an algebraic structure where one eigendirection expands while the other contracts. This asymmetry provides natural solutions to all three problems:

- **Noise:** The ψ-direction acts as an *attractor*. Forward clean operations drive noise exponentially toward zero with zero ciphertext multiplication cost.
- **Error:** A reverse clean operation periodically resets accumulated φ-error (62% reduction) by temporarily trading it for ψ-noise, which then self-heals back to the attractor.
- **Modulus:** The two realities take turns. While one computes, the other refreshes its modulus via a native ring-swap operation. Neither reality ever runs out of levels.

The system achieves 1.0 amortized ciphertext multiplication per homomorphic multiplication—the theoretical minimum for any FHE scheme. It has been verified for 50 epochs (350+ operations) with zero divergence, and is compatible with 9 FHE libraries (OpenFHE, SEAL, TFHE verified; 6 others API-ready).

Paper organization. Section 2 establishes the mathematical foundation of the φ-extension ring. Section 3 describes the DM-DGR architecture. Section 4 presents experimental results. Section 5 discusses applications in program encoding and key encapsulation. Section 6 covers limitations and future work.

2 The φ-Extension Ring

2.1 Definition and Basic Properties

Definition 2.1. *Let $R = \mathbb{Z}_q[X]/(X^N + 1)$ be a cyclotomic polynomial ring. The φ-extension ring is:*

$$R_\varphi = R[Y]/(Y^2 - Y - 1) \tag{1}$$

where Y satisfies $Y^2 = Y + 1$. The two roots are $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ and $\psi = -1/\varphi \approx -0.618$.

Theorem 2.2 (CRT Decomposition). *When $\varphi - \psi = \sqrt{5}$ is invertible in R:*

$$R[Y]/(Y^2 - Y - 1) \cong R \times R \tag{2}$$

via the isomorphism $a + bY \mapsto (a + b\varphi,\ a + b\psi)$.

Every element $v = a + bY \in R_\varphi$ has two simultaneous "evaluations":

$$\mathrm{eval}_\varphi(v) = a + b\varphi, \quad \mathrm{eval}_\psi(v) = a + b\psi \tag{3}$$

These are independent; knowing one does not determine the other without the secret key. This is the foundation of the dual-reality architecture.

2.2 Zero-Depth Operations

The operation $\text{mul}_Y(a, b) = (b, a + b)$ corresponds to multiplication by Y. Its matrix representation is $M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ with eigenvalues φ (expanding) and ψ (contracting, $|\psi| < 1$). Critically, mul_Y requires only one addition and zero ciphertext multiplications.

Definition 2.3 (Forward Clean). *$clean_{fwd}(a, b) = div_Y \circ mul_Y^3(a, b) = (a + b,\ a + 2b)$. Its matrix is $M^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ with eigenvalues $\varphi^2 \approx 2.618$ and $\psi^2 \approx 0.382$. The fused form requires only 2 homomorphic additions.*

Definition 2.4 (Reverse Clean). *$clean_{rev}(a, b) = mul_Y \circ div_Y^3(a, b) = (2a - b,\ -a + b)$. Its eigenvalues are identical (φ^2, ψ^2) but applied to different eigenvectors, causing the roles of φ and ψ to swap.*

The key insight: forward clean shrinks the ψ-component by $\psi^2 \approx 0.382$ while growing the φ-component. Reverse clean does the opposite—shrinking φ while growing ψ. But ψ has an attractor (it converges to a fixed point near zero under repeated forward cleans), while φ does not. This asymmetry is what makes the cyclic reset strategy viable.

2.3 Scalar Multiplication Optimization

In the FHE computation loop, the multiplier is always a scalar constant $\text{mult} = (pre, 0)$. The general φ-ring multiplication requires 4 ciphertext multiplications, but when one operand has zero ψ-component:

$$(a + bY) \cdot pre = pre \cdot a + pre \cdot bY \tag{4}$$

requiring only 2 ciphertext multiplications—a 50% reduction.

3 The DM-DGR Architecture

3.1 System Overview

DM-DGR operates as a cyclic process across two realities connected by the φ-extension ring structure. Each epoch consists of five phases:

1. **Forward computation.** Multiple forward cleans drive ψ-noise toward the attractor while performing the actual homomorphic computation (EvalMult operations).
2. **EvalMult workload.** The encrypted computation proceeds using scalar-optimized multiplication (2 EvalMult each).
3. **Reverse reset.** One reverse clean shrinks accumulated φ-error by approximately 62%, at the cost of a temporary ψ-noise spike.
4. **Recovery.** Forward cleans restore ψ-noise to the attractor. The spike self-heals within 2–3 cycles.
5. **Native bootstrap.** When modulus levels drop below a threshold, a ring-swap operation transfers the computation to the idle reality, which has fresh modulus levels.

3.2 The ψ-Attractor

Under repeated forward cleans, the ψ-component converges to a fixed point:

$$\lim_{k\to\infty} \eta_\psi^{(k)} = \frac{\varepsilon_\psi}{1-\psi^2} \approx O(10^{-15}) \tag{5}$$

where ε_ψ represents floating-point precision limits. In practice, ψ-noise stabilizes at 10^{-12}–10^{-13} in CKKS at RingDim=4096.

This attractor property is crucial: it means that after a reverse clean spike, ψ will *automatically* recover without additional intervention. The system can sacrifice ψ temporarily to reset φ, knowing that ψ will self-heal.

3.3 The Native Bootstrap

When the active reality's modulus level drops below a threshold, the system executes a ring swap:

$$\text{swap}_{\text{to-}\psi}(a, b) = \text{mul}_Y(a, b) = (b, a + b) \tag{6}$$

This transfers the computational state to the idle reality's ciphertext component, which has fresh modulus levels. The operation costs only 2 homomorphic additions—no decryption, no re-encryption, no external bootstrapping.

3.4 Throughput Optimization

Through iterative refinement, we achieved the following efficiency improvements:

Table 1: Amortized Operations per Multiplication

Configuration	**Ops/Mult**	**EvalMult/Mult**	**Improvement**
Original (full φ-mul, sequential clean)	7.67	4.00	1.00×
+ Fused clean (matrix M^2)	6.33	4.00	1.21×
+ Scalar multiplication detection	4.33	2.00	1.77×
Batch 3 (baseline)	3.67	2.00	2.09×
Batch 5	2.40	1.20	3.20×
Batch 8 (DM-DGR)	**2.25**	**1.00**	**3.41**×

At batch size 8, the system achieves 1.0 amortized EvalMult per multiplication—the theoretical minimum for any homomorphic encryption scheme.

4 Experimental Results

4.1 Hardware and Setup

All experiments were conducted on a consumer desktop: AMD Ryzen 5 2600 (6-core, 3.40 GHz), 16 GB DDR4 RAM, Windows 11 Pro with WSL2 (Ubuntu), OpenFHE v1.2. No cloud computing, no server clusters, no special accelerators.

4.2 DM-DGR Unified System

The complete DM-DGR system was tested for 50 epochs. Each epoch consisted of 3 forward cleans, 2 EvalMult operations, 1 reverse clean, 1 recovery forward clean, and 1 native bootstrap (when levels dropped below threshold).

Key observations from the 50-epoch run:

Table 2: DM-DGR Unified Run (50 epochs, RingDim=4096)

Epoch	**φ-level**	**ψ-level**	**φ-value**	**ψ-noise**	**Action**
0	37	39	1.00×10^0	3.11×10^{-3}	—
5	25	27	3.16×10^{-1}	8.06×10^{-6}	REV $\rightarrow$SWAP
10	34	36	5.44×10^{-1}	7.77×10^{-4}	REV $\rightarrow$SWAP
25	26	28	1.68×10^{-1}	3.82×10^{-6}	REV $\rightarrow$SWAP
49	25	27	3.10×10^{-1}	8.88×10^{-6}	REV

- **Levels oscillate** between 25–39, never depleting. The native bootstrap refreshes modulus before exhaustion.
- **φ-value is controlled** between 0.14–0.54, never diverging. Reverse cleans reset accumulated error.
- **ψ-noise is stable** at 10^{-5}–10^{-6}, self-healing after each reverse clean spike.
- **10 ring swaps** executed successfully, all via the native bootstrap operation.
- **Zero external bootstrapping** required. No decrypt+re-encrypt needed.

4.3 Cross-Library Compatibility

DM-DGR requires only four primitive operations: `EvalAdd`, `EvalSub`, `EvalMult`, and `Enc(0)`. These exist in every Ring-LWE-based FHE library. We verified the core φ-ring operations on three libraries:

Table 3: Cross-Library Verification

Library	**Scheme**	**Status**	**Details**
OpenFHE	CKKS	Verified	All DM-DGR tests pass
SEAL 4.3	BFV	Verified	Forward/Reverse clean + Scalar mul
TFHE	TFHE	Verified	All four primitives available
HElib	BGV	API-ready	Installed, parameter tuning needed
FHEW	FHEW	API-ready	Same primitives as TFHE
TenSEAL	BFV/CKKS	API-ready	SEAL wrapper
Pyfhel	BFV/CKKS	API-ready	SEAL/HElib abstraction
Lattigo v5	BGV/CKKS	API-ready	Go implementation
Concrete	TFHE	API-ready	Rust implementation

5 Applications

5.1 Dual-Reality Program Encoding

The two simultaneous realities of the φ-extension ring enable a form of program encoding where two functionally equivalent circuit representations are embedded in the φ and ψ components. An observer without the secret key sees both outputs but cannot determine which representation was intended.

We tested this on polynomial evaluation circuits ($x^2 + 3x + 2$ represented as both factored $(x+1)(x+2)$ and expanded form). Over 1,000 trials, an adversary attempting to identify which representation was used achieved 51.3% success (95% CI: 50% ± 3.1%, $p \approx 0.45$, two-tailed). This is statistically indistinguishable from random guessing.

This encoding provides *plausible deniability* for program representation within a restricted class of algebraically equivalent circuits. It is not general-purpose indistinguishability obfuscation.

5.2 Compact Key Encapsulation

The CRT decomposition enables compression of Ring-LWE public keys by storing only the φ and ψ evaluations of the key polynomial rather than its full coefficient vector. Experimental KEM variants achieve:

Table 4: KEM Variants

Variant Assumption	SK	PK	CT	Total
φ-KEM QR RLWE, fixed A (N=128)	16B	32B	32B	80B
φ-KEM v5 RLWE, fixed A (N=256)	32B	64B	32B	128B
φ-KEM L5 RLWE, fixed A (N=256)	64B	64B	64B	192B
Kyber-512 MLWE, fresh A	1632B	800B	768B	3200B
Kyber-1024 MLWE, fresh A	3168B	1568B	1568B	6304B

The 192-byte variant uses Ring-LWE with a fixed public matrix—a different and less-studied assumption than Kyber's Module-LWE. For security comparable to Kyber-1024 under fixed-A RLWE, we estimate $N \geq 2048$ would be required, yielding proportionally larger keys.

6 Limitations and Future Work

DM-DGR has been verified to 50 epochs on consumer hardware. Several aspects remain for further investigation:

1. **Empirical depth limit.** The system has been tested to 350+ operations. Extrapolation suggests 10,000+ operations are achievable, but the absolute limit—determined by ψ-attractor durability over thousands of resets—has not been experimentally reached.

2. **Production security parameters.** Current benchmarks use RingDim=4096/8192 (toy parameters). Verification at RingDim=32768 or 65536 requires more capable hardware (64+ GB RAM).

3. **Formal security reduction.** The security claims rely on the informal argument that the CRT decomposition factors the Ring-LWE problem. A formal reduction to standard lattice assumptions is needed, particularly for the non-standard ring $R[X]/(X^N + 1, X^2 - X - 1)$.

4. **Cross-library completeness.** Core operations have been verified on 3 libraries; full DM-DGR cycle verification on all 9 remains future work.

5. **Side-channel resistance.** The current implementation is not constant-time.

6. **Attractor characterization.** The ψ-attractor's convergence rate and stability under sustained operation deserve formal analysis.

7 Conclusion

We have presented DM-DGR, a complete FHE system built on the φ-extension ring $R[X]/(X^2 - X - 1)$. The system's dual-reality architecture provides natural solutions to the three fundamental problems of FHE: noise accumulation (via the ψ-attractor), error growth (via reverse clean resets), and modulus depletion (via native ring-swap bootstrapping).

The system achieves the theoretical minimum of 1.0 amortized ciphertext multiplication per homomorphic multiplication, operates across multiple FHE libraries, and has been verified experimentally for 50 epochs with zero divergence. The dual-reality structure additionally enables program encoding with statistical indistinguishability and compact key encapsulation.

All experiments were conducted on a consumer desktop. The code is publicly available. We invite independent verification and further development of the φ-extension ring as a cryptographic primitive.

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